COMPUTER GRAPHICS PROJECT

Bloxorz 3D

Three.js / WebGL puzzle game

SCOPE

What we built

8 verified levels
rolling cuboid
fragile tiles
heavy and soft switches
bridges
teleport split mode
level editor
saved progress
hint solver

Discrete first

```
state = x, z, orientation

orientation =
  standing
  lying_x
  lying_z

occupied cells decide:
  legal move
  fall
  switch
  goal
```

Animation follows the rule

```
input changes grid state

mesh position is interpolated
mesh rotation uses quaternion motion

after each move:
  snap back to exact grid coordinate
```

LEVEL MECHANICS

Rules on the board

normal tile	supports the block
fragile tile	breaks under standing block
heavy switch	standing block only
soft switch	any occupied cell
bridge	toggled tile
teleport	split into two cubes
goal	standing block only

Course connection

geometry
model transform
view transform
projection
rasterization
fragment shading
frame buffer

Scene setup

PerspectiveCamera
MeshPhysicalMaterial
DirectionalLight
AmbientLight
shadows
fog
SMAA
bloom
vignette
optional GPU path tracing

INTERACTION

Player input

keyboard controls
touch controls
orbit camera
move counter
par display
hints
localStorage progress

BFS over puzzle states

state key:
x, z
orientation
bridge mask
split status

verified par:

1	First Steps	8
2	Over the Edge	7
3	Fragile Ground	6
4	Open Sesame	6
5	Heavy Duty	12
6	Split Decision	8
7	Twin Bridges	13
8	The Gauntlet	15

CODE MAP

Files to explain

```
src/main.js         game flow
src/grid.js         board, tiles, bridges
src/controls.js     movement and validation
src/levels.js       levels, pars, themes
src/solver.js       hint search
src/editor.js       custom levels
src/scene.js        renderer, camera, lights
src/particles.js    visual feedback
scripts/check-levels.js   level verification
```

DEMO ORDER

Five minutes

1. level select and saved progress
2. Level 1 movement and goal
3. fragile tile
4. switch and bridge
5. teleport split and merge
6. hint
7. editor if time remains